4.13 Zeller's Congruence (Lab 4.2)

LAB ACTIVITY | 4.13.1: Zeller's Congruence (Lab 4.2) | Full screen | 0 / 40

Zeller's congruence is an algorithm developed by Christian Zeller to calculate the day of the week. The formula is:

$$h = \left(q + \left\lfloor \frac{(m+1)26}{10} \right\rfloor + K + \left\lfloor \frac{K}{4} \right\rfloor + \left\lfloor \frac{J}{4} \right\rfloor + 5J\right) \mod 7,$$

where:

- h is the day of the week (0 Sat, 1 Sun, 2 Mon, 3 Tues, 4 Wed, 5 Thur, 6 Fri)
- q is the day of the month
- m is the month (3:Mar, 4: Apr, 5:May ... 12:Dec). January and February are counted as months 13 and 14 of the previous year.
- J is the century (i.e. year//100)
- K is the year of the century (ie. year % 100)
- ⌊value⌋ indicates the "floor" of the given value (see Floor and ceiling functions).

While we could certainly use the math module to compute the floor values, we can instead use integer division to give us the same result (for positive values, as ours will be).

First, download this template and rename it `zeller.py`:

- template.py

Then, write a program called to compute the day of the week given a year, a month and a day of the month.

> **Prompt for and read** from the user values for the year, the month and the day of the month. Each of these values should be stored as integers.
> **If** the user entered 1 or 2 for the month it needs to be converted to 13 or 14 respectively, and the year should decrease by 1.
> **Compute** h using the formula above.
> **Create** a list of the days of the week. Start the list with Saturday, then Sunday, etc., ending with Friday
> **Print** the day of the week using h as an index for the days of the week list.

Here is an example execution.

```
Enter year: 2020↵
Enter month: 2↵
Enter day: 27↵
Thursday
```

Open new tab | Dock

Submit for grading

Coding trail of your work What is this?

```
History of your effort will appear here once you begin working on this zyLab.
```

Latest submission

No submissions yet

Template, Model Solution, and Test Bench Show

Lab statistics Show

Student behavior insights (beta) Show

Similarity detection Show

Student activity Show